Sparse Arrays and Cross-AP Position-Fused Estimation (CAPE) for Near-Field Cell-Free XL-MIMO: A Step-by-Step Framework

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1 Why we need a sparse array in our scenario

In a near-field cell-free system the useful benefit of an array grows with its *aperture* (physical width), not only with its antenna count. A *sparse* array spaces the same antennas wider, so for the same number of antennas we obtain a larger aperture. The concrete benefits in our scenario are:

- **Larger near-field region.** The near-field boundary (Rayleigh distance) is $d_R = 2D^2/\lambda$ [8, 7]; since it grows with the *square* of the aperture D, a wider sparse array turns far more links near-field, where distance-based (range) focusing becomes possible.
- **Same aperture with fewer antennas.** A sparse array reaches a target aperture with fewer elements than a dense one, i.e. fewer RF chains, which lowers hardware cost and the per-AP processing load [9, 3].
- **Sharper spatial resolution and localization**, because both improve with aperture [3, 6] (quantified in Sec. 2).

2 How the sparse array helps: conditions, resolution, localization

When it helps. The extra resolution matters only when users are hard to separate, i.e. *co-angular / crowded* users (close together, seen by an AP at nearly the same angle and distance) [6]. For well-separated users the network already resolves them and the gain is small.
Spatial resolution. The angular resolution scales as

$$\Delta s \;\sim\; \lambda/D, \tag{1}$$

Here: Here Δs is the smallest difference in the angle variable $s = \sin\theta$ that the array can still tell apart (two users closer than this look like one). λ is the wavelength (here 10 cm at 3 GHz), and D is the array width in metres. A *bigger* D makes Δs *smaller*, i.e. finer angle resolution. and near-field *range* focusing is possible only inside the beam-depth limit [3, Prop. 2]

$$r < r_{\text{th}} \approx \frac{d_R \cos^2\theta}{10.24}. \tag{2}$$

Here: Here r is how far the user is from the AP, and r_{th} is the distance below which the array can actually focus in range (tell a near user from a far user on the same line). θ is the user's angle (the $\cos^2\theta$ term says focusing weakens towards the array edges), and $d_R = 2D^2/\lambda$ is the Rayleigh distance. A larger aperture D raises d_R and hence r_{th}, so more users fall inside the range-focusing zone.

Localization. The position-error floor obeys the near-field Cramér–Rao scaling [8, 10]

$$\mathrm{std}(\hat{s}) \;\propto\; \frac{\lambda}{D\sqrt{p}}, \qquad \mathrm{std}(\hat{r}) \;\propto\; \frac{r^2}{D^2\sqrt{p}}. \tag{3}$$

Here: Here $\mathrm{std}(\hat{s})$ and $\mathrm{std}(\hat{r})$ are the typical errors in the estimated angle $\hat{s}$ and range $\hat{r}$. p is the signal-to-noise ratio (more power → smaller error), and D is the aperture. Both errors shrink as the aperture grows (the range error shrinks with D^2), which is why a wide sparse array localizes users far more accurately.

Why *non-uniform* sparse. A *uniform* sparse array (equal large spacing) creates strong ghost beams (grating lobes); a *non-uniform* sparse array such as the extended co-prime array (ECA) keeps the wide aperture but, through its irregular spacing, suppresses those ghosts, leaving only weak scattered side peaks [3, 4].

3 Proposed strategy (overview)

Using a sparse array *alone* gives the resolution of Sec. 2, but its wide spacing also produces *ambiguity*: besides the true focus the array has side peaks (for a uniform array, full-height grating lobes; for the ECA, weak ones). A single AP cannot always tell the true peak from a side peak, so per-AP estimation can lock onto the wrong one and the channel estimate degrades. To keep the resolution *and* remove the ambiguity we propose **CAPE** (Cross-AP Position-fused Estimation): several APs agree on the user's single physical position, which the side peaks of one AP do not support. We then show that CAPE is *complementary* to the cross-AP list-based successive interference cancellation (SIC) detector of [12, 13]: by supplying clean channels it makes that detector both more accurate and cheaper (it triggers its expensive all-AP fusion less often).

4 Sparse-array (ECA) model and antenna arrangement

There are L APs, each with a non-uniform sparse (ECA) array of N antennas, serving K single-antenna users on the uplink, all connected to a central processing unit (CPU) [1, 2].

Antenna arrangement (ECA). With co-prime integers $M, \tilde{N}$ (gcd = 1), one block is

$$\mathcal{B} = \{\, m\tilde{N} : 0 \le m < M \,\} \cup \{\, nM : 0 \le n < \tilde{N} \,\}, \tag{4}$$

Here: Here M and $\tilde{N}$ are two whole numbers with no common factor (we use 5 and 4). $\mathcal{B}$ is the set of antenna *positions* in one building block, measured in steps of $d_0 = \lambda/2$: five antennas spaced $\tilde{N}$ apart merged with four spaced M apart. The unequal gaps are what makes the array non-uniform. and the full array repeats the block B times and mirrors it,

$$\mathcal{N} = \bigcup_{b=0}^{B/2-1} \big\{ \pm\,(bM\tilde{N} + a) : a \in \mathcal{B} \big\}, \quad x_n = (\mathcal{N}_n - \bar{\mathcal{N}})\, d_0, \quad D = x_N - x_1. \tag{5}$$

Here: Here $\mathcal{N}$ is the complete list of antenna indices, obtained by placing B copies of the block end to end (period $M\tilde{N}$) and mirroring about the centre. x_n is the actual position of antenna n in metres ($\bar{\mathcal{N}}$ just centres the array), B is the number of blocks (even), $N = |\mathcal{N}|$ is the total antenna count, and D is the aperture (array width). For $M = 5, \tilde{N} = 4, B = 8$ this gives $N = 63$ antennas over $D = 7.6\,\mathrm{m}$.

Near-field channel. For a user at range $r_{l,k}$ and effective angle $s_{l,k}$ from AP l, the distance to antenna n and the (spherical-wave) channel are [5, 8]

$$r_{l,k,n} = \sqrt{r_{l,k}^2 + x_n^2 - 2\, r_{l,k}\, x_n\, s_{l,k}}\,, \tag{6}$$

$$\mathbf{h}_{l,k} = \sqrt{\beta_{l,k}}\; \mathbf{a}(r_{l,k}, s_{l,k}), \qquad [\mathbf{a}(r,s)]_n = \tfrac{1}{\xi}\tfrac{r}{r_n} e^{-\jmath\frac{2\pi}{\lambda}(r_n - r)}. \tag{7}$$

Here: Here $r_{l,k,n}$ is the exact distance from antenna n of AP l to user k (it differs across antennas because the wavefront is curved in the near field). $\mathbf{h}_{l,k}$ is the channel vector (one complex number per antenna). $\beta_{l,k}$ is the large-scale gain (path loss and shadowing, noise- normalized). $\mathbf{a}(r,s)$ is the *near-field steering vector*: the array's response to a source at range r and angle $s = \sin\varphi\cos\theta$; the term r/r_n is the near-field amplitude taper, the exponential is the phase, and ξ makes it unit length. The key point: $\mathbf{h}_{l,k}$ depends only on the user's *position* $(r_{l,k}, s_{l,k})$ — this is what CAPE exploits. A link is near-field when $r_{l,k} < d_R = 2D^2/\lambda$; otherwise it is far-field with covariance $\mathbf{R}_{l,k}$ [7].

Received pilots. With orthogonal pilots of length $\tau = K$, after de-spreading, AP l obtains for user k [1]

$$\mathbf{z}_{l,k} = \mathbf{h}_{l,k} + \mathbf{w}_{l,k}, \qquad \mathbf{w}_{l,k} \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_N), \quad \sigma^2 = \tfrac{1}{p\tau}. \tag{8}$$

Here: Here $\mathbf{z}_{l,k}$ is the noisy measurement AP l gets of user k's channel; $\mathbf{w}_{l,k}$ is additive Gaussian noise of power σ^2; p is the pilot SNR and $\tau = K$ the pilot length, so stronger pilots or more pilots give smaller σ^2. Each AP must estimate $\mathbf{h}_{l,k}$ from $\mathbf{z}_{l,k}$.

5 The ambiguity and the baseline (per-AP) estimator

The response between a trial point (r,s) and the true one is

$$g_l(r,s) = \left|\mathbf{a}(r,s)^{\mathsf{H}}\mathbf{a}(r_{l,k}, s_{l,k})\right| \in [0,1], \tag{9}$$

Here: Here $g_l(r,s)$ measures how much a beam aimed at the trial location (r,s) still "sees" the true user; $g = 1$ at the true spot. For a dense array g has one peak; for a sparse array it also has *side peaks* (the ambiguity). For a uniform sparse array these are strong grating lobes at $\sin\theta_u = \sin\theta_0 + 2u/\eta$ [3, Lemma 5]; for the ECA they are weak. The conventional estimator runs orthogonal matching pursuit (OMP) *independently* at each AP [11, 5]: it repeatedly picks the strongest peak and keeps it only if it clears a noise threshold,

$$(\hat{s}, \hat{q}) = \arg\max_{(s,q)} \left|\mathbf{a}(s,q)^{\mathsf{H}}\mathbf{r}\right|, \tag{10}$$

$$\text{accept iff} \quad \left|\mathbf{a}(\hat{s},\hat{q})^{\mathsf{H}}\mathbf{r}\right|^2 \geq \gamma\sigma^2, \quad \gamma = \ln D_{\text{dic}} + 0.58 + c, \tag{11}$$

Here: Here $(\hat{s},\hat{q})$ is the estimated angle $\hat{s}$ and inverse range $\hat{q} = 1/\hat{r}$ of one path; $\mathbf{r}$ is the leftover signal after removing earlier paths (the residual). Equation (11) rejects noise-only peaks: $\gamma\sigma^2$ is the level the strongest of D_{dic} pure-noise correlations reaches (D_{dic} is the number of trial atoms, c a safety margin). **Problem on a sparse array:** on a weak link, (10) may pick a *side peak* of g_l, so the estimated channel points the wrong way. The channel then uses an LMMSE (Wiener) weighting of the selected atoms, giving $\hat{\mathbf{h}}^{\text{loc}}_{l,k}$ and error variance $\varepsilon^2_{l,k}$, with $\mathbf{C}_{l,k} = \varepsilon^2_{l,k}\mathbf{I}_N$ [5].

6 CAPE: cross-AP position-fused estimation (proposed)

Idea. A user has one physical location $\mathbf{u}_k$, and by (7) every AP's channel is a known function of it. We find the single location the strongest APs agree on; a side peak of one AP is rejected because the other APs do not support it.

Step 1 — geometry of a trial location. For any trial point $\mathbf{u} \in \mathbb{C}$,

$$d_l(\mathbf{u}) = |\mathbf{u} - \mathbf{c}_l|, \quad r_l(\mathbf{u}) = \sqrt{\Delta h^2 + d_l^2(\mathbf{u})}, \quad s_l(\mathbf{u}) = \sin\big(\angle(\mathbf{u} - \mathbf{c}_l)\big)\tfrac{d_l(\mathbf{u})}{r_l(\mathbf{u})}, \tag{12}$$

Here: Here $\mathbf{u}$ is any guessed user location on the map, $\mathbf{c}_l$ is AP l's location, d_l is the flat (horizontal) distance between them, Δh is the AP–user height difference, r_l is the true 3-D distance, and s_l is the matching effective angle. From these, AP l's expected atom is $\mathbf{a}_l(\mathbf{u}) = \mathbf{a}(r_l(\mathbf{u}), s_l(\mathbf{u}))$. In words: pick a point on the map and every AP knows which beam would point there.

Step 2 — candidate locations. Each strong local peak $(\hat{s}, \hat{q})$ of the N_{c} strongest APs maps to two mirror locations (a line array cannot tell front from back):

$$\hat{r} = 1/\hat{q}, \;\; d = \sqrt{\hat{r}^2 - \Delta h^2}, \;\; \hat{\varphi} \in \Big\{ \arcsin\tfrac{\hat{s}\hat{r}}{d}, \, \pi - \arcsin\tfrac{\hat{s}\hat{r}}{d} \Big\}, \;\; \mathbf{u} = \mathbf{c}_l + d\, e^{\jmath\hat{\varphi}}. \tag{13}$$

Here: Here each AP turns its own peak into map points: $\hat{r} = 1/\hat{q}$ is the guessed range, d the horizontal distance, and $\hat{\varphi}$ the azimuth — there are *two* because the array cannot distinguish a point in front from its mirror. Collecting these from the N_{c} strongest APs gives the candidate set $\mathcal{X}_k$ (the true location plus several false ones).

Step 3 — cross-AP fused likelihood. Score each candidate by how well it matches the N_{AP} strongest APs *together* (each AP has an unknown gain, hence magnitudes):

$$J_k(\mathbf{u}) = \sum_{l \in \mathcal{F}_k} \left| \mathbf{a}_l(\mathbf{u})^{\mathsf{H}} \mathbf{z}_{l,k} \right|^2, \qquad \hat{\mathbf{u}}_k = \arg\max_{\mathbf{u} \in \mathcal{X}_k} J_k(\mathbf{u}). \tag{14}$$

Here: Here $J_k(\mathbf{u})$ is the total "agreement" score of a candidate location across the fused AP set $\mathcal{F}_k$ (the N_{AP} strongest APs): each term measures how strongly AP l's measurement $\mathbf{z}_{l,k}$ supports the beam $\mathbf{a}_l(\mathbf{u})$. The true location scores high at all APs; $\hat{\mathbf{u}}_k$ is the best-scoring candidate.

Step 4 — refinement. Polish $\hat{\mathbf{u}}_k$ by a local 2-D pattern search (step δ, halved on failure):

$$\hat{\mathbf{u}}_k \leftarrow \arg\max_{\mathbf{u} \in \{\hat{\mathbf{u}}_k + \delta e^{\jmath 2\pi i/8}\}_{i=0}^{7} \cup \{\hat{\mathbf{u}}_k\}} J_k(\mathbf{u}), \qquad \delta \leftarrow \delta/2. \tag{15}$$

Here: Here δ is the step of a small search around the current estimate: we test eight neighbours; if one is better we move there, otherwise we halve δ and look closer. This fine-tunes the position to sub-metre accuracy.

Step 5 — per-AP channel from the agreed position, with a gate. For every near-field AP l, build the atom from $\hat{\mathbf{u}}_k$ ($\mathbf{b}_l^{\mathrm{fus}}$) and keep the AP's own local atom $\mathbf{b}_l^{\mathrm{loc}}$ only when it is clearly better:

$$\mathbf{b}_l = \begin{cases} \mathbf{b}_l^{\mathrm{loc}}, & |\mathbf{b}_l^{\mathrm{loc}\,\mathsf{H}} \mathbf{b}_l^{\mathrm{fus}}| \geq \mu_{\mathrm{c}} \text{ and } m_l^{\mathrm{loc}} > m_l^{\mathrm{fus}}, \text{ or } (m_l^{\mathrm{loc}})^2 - (m_l^{\mathrm{fus}})^2 > \gamma_0 \sigma^2, \\ \mathbf{b}_l^{\mathrm{fus}}, & \text{otherwise}, \end{cases} \tag{16}$$

Here: Here $\mathbf{b}_l$ is the atom AP l finally uses. $\mathbf{b}_l^{\mathrm{fus}}$ is built from the agreed position, $\mathbf{b}_l^{\mathrm{loc}}$ is the AP's own best local atom, and $m_l^{(\cdot)} = |\mathbf{b}_l^{(\cdot)\mathsf{H}} \mathbf{z}_{l,k}|$ is how well each fits the data. The gate keeps the local one only if it sits on the same lobe (μ_{c} is the overlap threshold) and fits better, or fits better than the noise level $\gamma_0 \sigma^2$. This guarantees CAPE is never worse than local estimation at high SNR. The channel uses a prior Wiener gain and a residual-based error variance,

$$\hat{\mathbf{h}}_{l,k}^{\mathrm{cape}} = \mathbf{b}_l \, \frac{N\beta_{l,k}}{N\beta_{l,k} + \sigma^2} \, (\mathbf{b}_l^{\mathsf{H}} \mathbf{z}_{l,k}), \qquad \mathbf{C}_{l,k} = \varepsilon_{l,k}^2 \mathbf{I}_N. \tag{17}$$

Here: Here $\hat{\mathbf{h}}_{l,k}^{\mathrm{cape}}$ is the final channel estimate: the direction $\mathbf{b}_l$ scaled by the measured amplitude $\mathbf{b}_l^{\mathsf{H}} \mathbf{z}_{l,k}$ and a Wiener gain $N\beta_{l,k}/(N\beta_{l,k} + \sigma^2)$ that shrinks weak/noisy links. $\varepsilon_{l,k}^2$ is the leftover error power (from the residual), and $\mathbf{C}_{l,k}$ the error covariance used by the detector. Far-field links keep the statistical LMMSE estimate [7].

Why fusion removes the ambiguity. For the true location $\mathbf{u}_k$ versus a false one $\mathbf{u}'$ that is a side peak of AP $l^\star$ with level $\rho = g_{l^\star}(\mathbf{u}')$, using $\|\mathbf{h}_{l,k}\|^2 \approx N\beta_{l,k}$,

$$\mathbb{E}[J_k(\mathbf{u}_k)] \approx \sum_{l \in \mathcal{F}_k} (N\beta_{l,k} + \sigma^2), \tag{18}$$

$$\mathbb{E}[J_k(\mathbf{u}')] \approx N\beta_{l^\star,k}\rho^2 + \sum_{m \in \mathcal{F}_k \setminus \{l^\star\}} (N\beta_{m,k}\rho_m^2 + \sigma^2), \quad \rho_m \ll 1, \tag{19}$$

$$\frac{\mathbb{E}[J_k(\mathbf{u}_k)]}{\mathbb{E}[J_k(\mathbf{u}')]} \gtrsim \frac{\sum_{l \in \mathcal{F}_k} N\beta_{l,k}}{N\beta_{l^\star,k}\rho^2} \xrightarrow[|\mathcal{F}_k| \text{ large}]{} \gg 1. \tag{20}$$

Here: Here $\rho = g_{l^\star}(\mathbf{u}')$ is how strong the false point looks at the one AP ($l^\star$) that has a side peak there, while $\rho_m \ll 1$ is how weak it looks at every other AP m (a different AP sees that point at a different angle/range, so it does not line up). Thus the true location's score (18) adds up over *all* APs, but the false one (19) is supported by only one AP; the ratio (20) grows with the number of fused APs $|\mathcal{F}_k|$. This needs $\rho_m \ll 1$, which holds for the ECA (irregular side peaks) but *not* for a uniform sparse array (strong, regular grating lobes).

7 Detection and the benefit of CAPE to the cross-AP list detector

Each AP forms a local MMSE combiner and the CPU fuses the AP outputs by large-scale fading decoding (LSFD) [2]; hard symbols are resolved by the cross-AP list-based SIC detector of our prior work [12, 13]:

$$\mathbf{v}_{l,k} = \Big(p\,\hat{\mathbf{H}}_l\hat{\mathbf{H}}_l^{\mathsf{H}} + p\sum_j \mathbf{C}_{l,j} + \mathbf{I}_N\Big)^{-1}\hat{\mathbf{h}}_{l,k}, \qquad \bar{B}_k = |\mathcal{S}_k| + \rho\,(L - |\mathcal{S}_k|). \tag{21}$$

Here: Here $\mathbf{v}_{l,k}$ is AP l's receive filter for user k (built from the estimated channels $\hat{\mathbf{H}}_l$ and their error covariances $\mathbf{C}_{l,j}$). $\mathcal{S}_k$ is the set of APs serving user k (its cluster), L the total APs, and $\bar{B}_k$ the fronthaul cost: the cluster always sends $|\mathcal{S}_k|$ values, plus the extra all-AP fusion for the remaining $L - |\mathcal{S}_k|$ APs is used only a fraction ρ of the time (the gate-firing rate).
Role of CAPE for the detectors. From the measured results we observe:

- **Simple / list detectors (Linear, hard-SIC, list-SIC):** these use *per-AP* resolution, so the sparse array helps them *even without* CAPE (e.g. list-SIC BER $\approx 2.1 \times 10^{-2}$ for ECA vs 3.1×10^{-2} for the dense array at 20 dB, crowded).
- **Cross-AP list detector *without* CAPE:** fusing *inaccurate* ECA estimates across APs can propagate errors, so the plain ECA can be *worse* than the dense array (1.7×10^{-3} vs 1.3×10^{-3}).
- **Cross-AP list detector *with* CAPE:** clean channels make the fusion reliable, giving the best end-to-end BER (5.5×10^{-4}, $\approx 2.3\times$ below the dense array) *and* a lower gate-firing rate ρ, i.e. *lower fronthaul / complexity.*

Hence CAPE and the cross-AP list detector are complementary: ρ in (21) drops because better channels reduce the hard cases, so CAPE both improves accuracy and reduces the detector's cost. The gains are large for crowded/co-angular users and shrink to a tie for well-separated users.

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